

# CareAIFlow – Standardized Compliance Folder System with Smart Monitoring (Phase 2)

*(Compliance Visibility, Document Readiness & Operational Intelligence Layer)*

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## 1. Executive Overview

This phase introduces a structured **Compliance Visibility and Monitoring Layer** designed to enhance the effectiveness of the existing document management system within CareAIFlow.

The current system already provides:

- Folder-based document organization
- Automatic folder creation
- Document upload and storage capabilities
- Flexible categorization and tagging

While these features enable document storage, they rely heavily on **manual organization and user awareness**, which creates operational challenges such as:

- Inconsistent document completeness across profiles
  - Lack of visibility into missing or incomplete records
  - Time-consuming manual verification
  - Increased risk during inspections and audits
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## Purpose of This Phase

This enhancement introduces a system that actively **monitors, evaluates, and communicates document completeness**, allowing users to:

- Instantly identify missing documentation
  - Understand compliance status at a glance
  - Receive guidance on required actions
  - Maintain inspection-ready records without manual checking
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## System Transition

This phase transforms the platform from:

Passive document storage

to

**Active compliance visibility and document readiness system**

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## **2. Strategic Objectives**

The key objectives of this phase are:

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### **2.1 Improve Operational Visibility**

Provide clear, real-time insights into document completeness across all profiles.

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### **2.2 Reduce Manual Oversight**

Eliminate the need for manual checking of folders by introducing automated detection and status tracking.

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### **2.3 Support Compliance Readiness**

Ensure that documentation is consistently organized and complete, reducing audit risk.

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### **2.4 Standardize Compliance Awareness**

Introduce a uniform way of understanding document completeness across residents, staff, and facility records.

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### **2.5 Enhance User Efficiency**

Enable staff to quickly identify what needs attention without navigating multiple folders.

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## **3. Scope Coverage**

This system applies to the following entities:

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### 3.1 Resident Profiles

- Admission documentation
  - Medical records
  - Consent and legal documents
  - Care-related documentation
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### 3.2 Staff Profiles

- Certifications and licenses
  - Compliance documents
  - Training records
  - Employment documentation
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### 3.3 Facility Records

- Licensing and regulatory documents
  - Operational logs
  - Policy and compliance records
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## 4. System Architecture Overview

This phase introduces a **monitoring layer that operates on top of the existing folder structure**.

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### 4.1 Layer-Based Design

#### Layer 1 – Existing System (Already Implemented)

- Folder structure
  - Document storage
  - Upload functionality
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#### Layer 2 – Compliance Monitoring Layer (This Phase)

This new layer introduces:

- Continuous evaluation of document completeness
  - Status calculation per category
  - Identification of missing or incomplete data
  - Real-time feedback and indicators
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## **4.2 Data Evaluation Logic**

The system will evaluate:

- Presence of documents within predefined categories
  - Completeness of required document groups
  - Status of each category (Complete / Partial / Missing)
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## **4.3 Output Channels**

The monitoring system feeds into:

- Profile-level views
  - Folder-level indicators
  - Dashboard summaries
  - Real-time UI feedback
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# **5. Core Feature Set – Deep Dive**

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## **5.1 Missing Document Detection System**

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### **Overview**

A core engine that continuously scans document categories and identifies gaps in documentation.

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### **Functional Behavior**

The system evaluates each predefined folder category and determines:

- Whether required documents exist
  - Whether document sets are complete
  - Whether any expected categories are empty
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## **Detection Types**

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### **A. Missing Documents**

- Entire category has no documents

Example:

- No consent forms uploaded
  - No medication-related documentation
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### **B. Incomplete Document Sets**

- Partial documentation exists but not complete

Example:

- Admission document present but missing supporting forms
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### **C. Category Gaps**

- Required compliance categories are not populated
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## **System Output**

- Flagged missing categories
  - Highlighted incomplete sections
  - Structured list of missing items
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## **Behavior Characteristics**

- Runs automatically in the background
- Updates dynamically after uploads

- Requires no manual triggering
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## 5.2 Compliance Status Monitoring Engine

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### Overview

Provides a structured view of document completeness for each entity.

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### Status Calculation Model

Each category is evaluated and assigned a status:

- **Complete** → All required documents present
  - **Partial** → Some documents present, but incomplete
  - **Missing** → No documents present
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### Aggregation Logic

The system calculates:

- Category-level status
  - Profile-level overall status
  - Summary of missing or incomplete items
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### Example Output

Resident Profile:

| Category       | Status   |
|----------------|----------|
| Identification | Complete |
| Consents       | Missing  |
| Medications    | Partial  |

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### Profile-Level Indicators

- Overall completion state
- Quick summary of missing areas
- Readiness visibility

# Intelligence Layer, UI System & User Interaction Design

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## 6. AI Recommendations Engine (Structured Intelligence Layer)

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### 6.1 Overview

The AI Recommendations Engine provides **context-aware, actionable guidance** to users based on document completeness and compliance status.

This is not a complex autonomous AI system, but a **controlled, rule-driven intelligence layer** designed to:

- Guide users toward completion
  - Reduce decision-making effort
  - Improve overall system usability
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### 6.2 Core Functionality

The system analyzes:

- Missing document categories
- Incomplete document sets
- Profile-level compliance gaps

And generates **targeted recommendations**.

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### 6.3 Recommendation Types

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#### A. Missing Document Prompts

Generated when a required category has no documents.

**Examples:**

- “Upload consent form”
  - “Add admission documentation”
  - “Upload medication order”
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**B. Incomplete Record Prompts**

Triggered when partial documentation exists.

**Examples:**

- “Complete admission records”
  - “Add remaining documents for care planning”
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**C. General Completion Guidance**

High-level recommendations based on overall profile status.

**Examples:**

- “Complete remaining compliance documents”
  - “Review incomplete sections”
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**6.4 Delivery Mechanism**

Recommendations will be displayed:

- Within profile views
  - Inside dashboard summaries
  - Alongside folder categories
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**6.5 Behavior Design**

- Updates dynamically based on document activity
  - Prioritizes most critical gaps
  - Avoids overloading the user with excessive suggestions
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## 7. Compliance Dashboard System

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### 7.1 Overview

A centralized dashboard providing **real-time visibility into document completeness and compliance readiness**.

This acts as the **primary control center** for monitoring documentation across the platform.

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### 7.2 Dashboard Components

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#### A. Summary Overview Panel

Displays:

- Overall compliance status
  - Total profiles monitored
  - Number of incomplete profiles
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#### B. Category-Level Breakdown

Shows:

- Completion status by category
  - Distribution of Complete / Partial / Missing
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#### C. Missing Items Panel

Highlights:

- Critical missing documents
  - Profiles with highest gaps
  - Actionable items requiring attention
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## D. Quick Navigation

Allows users to:

- Jump directly to incomplete profiles
  - Access missing document categories
  - Resolve issues quickly
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## 7.3 Data Representation

- Visual indicators (status badges, color-coded states)
  - Simple progress-style indicators
  - Clean and readable layout
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## 7.4 Use Cases

- Daily operational monitoring
  - Compliance tracking
  - Pre-audit preparation
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# 8. Real-Time Feedback & Alert System

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## 8.1 Overview

Provides immediate visual and contextual feedback to users as they interact with the system.

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## 8.2 Feedback Types

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### A. Status Indicators

Each category and profile will display:

- **Complete** (all documents present)
- **Partial** (incomplete documentation)

- **Missing** (no documents available)
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## **B. Visual Highlighting**

- Missing categories highlighted
  - Incomplete sections marked clearly
  - Priority gaps visually emphasized
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## **C. Contextual Alerts**

Displayed when:

- Required documents are missing
  - Incomplete categories are accessed
  - Users attempt to proceed with gaps
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## **8.3 Behavior Characteristics**

- Updates instantly after document upload
  - Reflects real-time system state
  - Requires no manual refresh
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# **9. User Interaction Flow (End-to-End)**

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## **9.1 Profile-Level Interaction**

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### **Step 1: User Opens Profile**

- Sees overall compliance status
  - Identifies missing categories immediately
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### **Step 2: User Navigates to Folder**

- Views category-level status
  - Sees missing or incomplete indicators
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### **Step 3: System Suggests Actions**

- Displays recommendations
  - Guides user to upload required documents
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### **Step 4: User Uploads Document**

- Status updates automatically
  - Missing indicators adjust in real time
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### **Step 5: Profile Status Updates**

- Compliance improves
  - Dashboard reflects change instantly
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## **9.2 Dashboard Interaction**

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### **Step 1: User Accesses Dashboard**

- Views system-wide compliance overview
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### **Step 2: Identifies Issues**

- Sees incomplete profiles
  - Identifies high-priority gaps
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### **Step 3: Navigates to Profile**

- Direct access to problem areas

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#### **Step 4: Resolves Issues**

- Uploads documents
  - Completes categories
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## **10. System Behavior Design**

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### **10.1 Automation Level**

- Fully automatic evaluation
  - No manual trigger required
  - Continuous monitoring
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### **10.2 Performance Considerations**

- Lightweight background processing
  - Efficient data checks
  - Minimal system load impact
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### **10.3 Scalability**

Designed to support:

- Increasing number of users
- Larger document sets
- Future compliance features

## **System Logic, Implementation Depth & Final Positioning**

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## **11. Data Structure & Evaluation Logic**

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## 11.1 Category-Based Evaluation Model

Each entity (Resident, Staff, Facility) will be evaluated based on:

- Predefined document categories (from standardized folder structure)
  - Presence or absence of documents within each category
  - Completeness of document sets within each category
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## 11.2 Evaluation Layers

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### Layer 1 – Presence Check

- Does the category contain any document?
  - Output: Present / Missing
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### Layer 2 – Completeness Check

- Does the category meet minimum expected documentation?
  - Output: Complete / Partial
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### Layer 3 – Aggregated Status

- Combines all category statuses into:
    - Overall profile status
    - Summary of missing areas
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## 11.3 Status Calculation Framework

Each category will follow:

- **Complete** → Required documents present
  - **Partial** → Some documents present
  - **Missing** → No documents present
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## 11.4 Profile-Level Aggregation

The system will compute:

- Total categories evaluated
- Number of completed categories
- Number of missing/incomplete categories

And generate:

- Overall compliance status
  - Missing item summary
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## 12. Real-Time Processing Logic

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### 12.1 Event-Based Updates

The system recalculates status when:

- A document is uploaded
  - A document is deleted
  - A document is updated
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### 12.2 Background Monitoring

- Continuous lightweight checks
  - Ensures dashboard and profile data remain accurate
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### 12.3 Sync Behavior

- Profile view, dashboard, and folder view remain synchronized
  - Changes reflect instantly across all views
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## 13. Edge Case Handling

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## 13.1 Empty Profiles

- Profiles with no documents will show:
    - All categories as “Missing”
    - Full recommendation list
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## 13.2 Partial Upload Scenarios

- When only some documents are uploaded:
    - Category marked as “Partial”
    - Suggestions generated accordingly
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## 13.3 Multiple Documents in Category

- System does not limit uploads
  - Status remains based on completeness, not quantity
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## 13.4 Optional Categories

- Some categories may not apply to all profiles
  - System allows flexibility without breaking status logic
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# 14. Integration with Existing System

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## 14.1 Compatibility

This system will:

- Fully integrate with existing folder structure
  - Use current document storage
  - Require no migration
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## **14.2 Non-Disruptive Deployment**

- No changes to current workflows
  - Users continue using system as-is
  - New features enhance visibility only
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## **14.3 Data Integrity**

- Existing documents remain unchanged
  - No data restructuring required
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# **15. System Performance & Scalability**

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## **15.1 Performance Design**

- Lightweight evaluation logic
  - Optimized for fast response
  - Minimal impact on system performance
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## **15.2 Scalability**

Designed to support:

- Growth in number of users
  - Increased document volume
  - Expansion to additional modules
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# **16. Security & Access Considerations**

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## **16.1 Role-Based Visibility**

- Compliance status visible based on user roles
  - Sensitive data access remains controlled
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## **16.2 Data Protection**

- No changes to existing security protocols
  - Document access remains permission-based
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# **17. Implementation Approach**

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## **17.1 Phase-Based Execution**

Implementation will follow:

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### **Step 1 – Evaluation Engine Setup**

- Category mapping
  - Status logic implementation
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### **Step 2 – Profile-Level Integration**

- Status indicators in profiles
  - Recommendation engine activation
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### **Step 3 – Dashboard Development**

- System-wide compliance view
  - Missing item tracking
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### **Step 4 – Real-Time Feedback Integration**

- Alerts
- Visual indicators

- Status updates
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## 17.2 Validation & Testing

- Test across all entity types
  - Validate status accuracy
  - Ensure UI consistency
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## 18. Scope Boundaries (Reinforced)

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### Included

- ✓ Missing document detection
  - ✓ Compliance status monitoring
  - ✓ Recommendation engine
  - ✓ Dashboard system
  - ✓ Real-time feedback
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### Not Included

- ✗ AI document classification
  - ✗ Cross-module validation
  - ✗ Inspection simulation
  - ✗ Predictive analytics
  - ✗ External integrations
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## 19. Final Outcome

After this phase, CareAIFlow will provide:

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### Operational Benefits

- Clear visibility into document completeness
  - Faster identification of missing items
  - Reduced manual effort
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## **Compliance Benefits**

- Improved audit readiness
  - Structured documentation visibility
  - Reduced compliance risk
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## **User Experience Benefits**

- Guided document management
  - Simplified workflows
  - Better system usability
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## **20. Final Positioning**

This phase is not:

- ✗ A folder enhancement
  - ✗ A UI improvement
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This phase is:

**A Compliance Visibility & Document Readiness System**

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## **21. Strategic Value**

This enhancement establishes:

- A foundation for compliance automation
- A scalable monitoring system
- A differentiated product capability